

# MIT OpenCourseWare

6.854J

Advanced Algorithms 6.854J — Advanced Algorithms May 2026 **EXTREMELY****HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (7 points) Prove correctness of the Cheney garbage collection algorithm using a formal argument about the copy phase preserving the reachability graph.
2. (2 points) Prove the parallel repetition theorem for interactive proofs: if a verifier accepts with probability  $p$  in a single round, then the verifier accepts in  $n$  parallel repetitions with probability at most  $p^{\Omega(n)}$  (for certain protocols).
3. (3 points) Prove that the matrix multiplication exponent  $\omega$  satisfies  $2 \leq \omega < 2.373$  by analyzing the Coppersmith-Winograd algorithm and its subsequent improvements.
4. (5 points) Prove the time hierarchy theorem: for time-constructible functions  $f(n)$ , there exists a language decidable in  $O(f(n))$  time but not in  $o(f(n)/\log f(n))$  time.
5. (4 points) Prove the Goldreich-Levin theorem: if  $f$  is a one-way function, then there exists a one-way function  $g$  and a hardcore predicate  $b$  such that  $b(x)$  is hard to predict given  $g(x)$ .
6. (3 points) Prove the Karger-Stein randomized min-cut algorithm runs in  $O(n^2 \log^3 n)$  time and succeeds with high probability. Analyze the probability that a particular min-cut survives the contraction process.
7. (4 points) Prove the Byzantine Generals Problem has no solution for  $n \leq 3m$  generals with  $m$  traitors in the oral messages model. Show the  $n \geq 3m+1$  necessary condition.
8. (3 points) Prove the Adleman theorem:  $\text{BPP} \subseteq \text{P/poly}$ . Show that any randomized polynomial-time algorithm can be derandomized with polynomial-size advice.
9. (2 points) Prove that 3-SAT reduces to Hamiltonian Cycle in polynomial time. Construct the explicit gadget for each clause and variable, showing that a satisfying assignment corresponds to a Hamiltonian cycle in the constructed graph.
10. (3 points) Prove that  $\text{EXP}^{\text{NP}}$  contains languages that require exponential-size circuits (Karp-Lipton theorem variant). Show that if  $\text{NP} \subseteq \text{P/poly}$  then  $\text{PH} = \Sigma_2^P$ .

11. (5 points) Prove the CAP theorem: a distributed data store cannot simultaneously provide more than two of Consistency, Availability, and Partition tolerance. Formalize the theorem using a proof by contradiction with atomic registers.
12. (3 points) Prove that parity cannot be computed by a polynomial-size constant-depth circuit ( $AC^0$ ) using the switching lemma. Show that any  $AC^0$  circuit computing parity must have size  $\exp(\Omega(n^{1/(d-1)}))$ .
13. (3 points) Prove that problem  $RL \subseteq SC$  (randomized logspace is in Steve's class) using the Nisan generator and showing that  $RL$  can be derandomized with polynomial seed length.
14. (5 points) Prove the PCP theorem:  $NP = PCP(\log n, 1)$ . Construct a verifier for SAT that reads  $O(1)$  bits of the proof and uses  $O(\log n)$  random bits.
15. (5 points) Prove that if  $P = NP$ , then  $BPP = P$ . Conversely, show that  $BPP \subseteq EXP$  and discuss the relationship between  $P$  and  $BPP$ .
16. (9 points) Prove that one-way functions exist if and only if  $P \neq NP$ . Construct a one-way function based on the hardness of SAT, or show that the existence of OWFs is equivalent to pseudorandom generators.
17. (2 points) Prove that graph isomorphism is in  $NP \cap coAM$ . Show that GI is unlikely to be NP-complete (if it were, PH collapses).
18. (6 points) Prove the VC-dimension bound for PAC learning: if a hypothesis class  $\mathcal{H}$  has VC-dimension  $d$ , then the sample complexity for  $(\epsilon, \delta)$ -PAC learning is  $m \geq \Omega(\frac{1}{\epsilon}(d + \log(1/\delta)))$ . Provide the upper bound as well.
19. (4 points) Prove type soundness for the simply typed lambda calculus using the progress and preservation lemmas. Extend the proof to System F with parametric polymorphism.
20. (3 points) Prove that the minimum distance of a linear code can be computed using the parity check matrix and show that the decisional version is NP-complete (Berlekamp-Welch theorem variant).
21. (5 points) Prove that the amortized cost of a sequence of  $n$  operations in a Fibonacci heap is  $O(\log n)$  for extract-min and  $O(1)$  for all other operations, using the potential function method.
22. (3 points) Prove that directed Steiner tree is NP-hard by reducing from set cover. Then show it is approximable within  $O(\log^2 n)$ .

23. (3 points) Prove the Chernoff bound: for independent Bernoulli random variables  $X_1, \dots, X_n$  with  $\Pr[X_i = 1] = p_i$ , let  $X = \sum X_i$  and  $\mu = \mathbb{E}[X]$ . Show that  $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1+\delta)^{1+\delta}})^\mu$  for all  $\delta > 0$ .
24. (3 points) Prove the Cook-Levin theorem: SAT is NP-complete. Construct a polynomial-time reduction from any NP language  $L$  to SAT.
25. (5 points) Prove an  $\Omega(n \log n)$  lower bound for sorting in the comparison-based model using the adversary method. Then apply the adversary argument to prove a lower bound for finding the  $k$ -th smallest element.

## Solutions

### Solution 1:

Cheney's algorithm uses a semi-space collector. Maintain *fromspace* and *tospace*. The scan pointer iterates through *tospace* while the alloc pointer advances. Invariant: all reachable objects from roots have been copied or are in the queue. Each edge traversed preserves connectivity:  $\forall x$  reachable from root,  $x$  is either in *tospace* (scanned or unscanned) or reachable from unscanned objects. Termination: when scan = free, all reachable objects are in *tospace*.

### Solution 2:

For XOR games, Raz's parallel repetition theorem shows the value decays exponentially:  $\omega(G^{\otimes n}) \leq \omega(G)^{cn}$  where  $c$  depends on the game. The proof uses information theory: the winning probability for each coordinate is correlated only through the initial shared randomness, and the protocol cannot coordinate between coordinates without increasing communication.

### Solution 3:

Strassen's  $O(n^{2.807})$  algorithm uses the identity  $M_2(2) = 7$  giving  $\omega = \log_2 7$ . Coppersmith-Winograd uses the laser method and bilinear algebra to achieve  $\omega < 2.376$ . The key idea: construct a family of tensors  $T_n$  with border rank  $r_n$  and combining them recursively. The asymptotic inequality  $\omega < \log_{n^{2/3}}(r_n)$  yields improved bounds. The current best  $\omega < 2.37286$  uses refined laser method analysis.

### Solution 4:

Construct a diagonalizing TM  $D$  that on input  $x$  runs  $M_x$  on  $x$  for  $f(|x|)$  steps and outputs the opposite answer. If  $D$  runs in  $o(f(n)/\log f(n))$  time, we get a contradiction for all sufficiently large  $x$ . The  $\log f(n)$  overhead comes from simulation costs:  $D$  must simulate  $M_x$  which has a different tape alphabet and number of tapes.

### Solution 5:

Define  $g(x, r) = (f(x), r)$  for  $|x| = |r|$ . The hardcore predicate  $b(x, r) = x \cdot r \pmod 2$ . If an adversary predicts  $b$  with advantage  $\varepsilon$ , construct an inverter for  $f$ : use a clever reconstruction (Goldreich-Levin list decoding) that, given  $f(x)$  and oracle access to the predictor, outputs a list containing  $x$  with high probability. The reconstruction uses pairwise independent sampling and majority decoding.

### Solution 6:

Contract edges uniformly until  $t$  vertices remain. A fixed min-cut  $C$  of size  $k$  survives a single contraction step with probability  $\geq 1 - 2/t$ . By induction, survival probability  $\geq \binom{t}{2}/\binom{n}{2}$ . Setting  $t = n/\sqrt{2}$  and recursing twice gives  $O(n^2 \log n)$  expected time. The success probability is  $\Omega(1/\log n)$ , and repeating  $O(\log^2 n)$  times gives high probability.

### Solution 7:

For  $m = 1, n = 3$ : assume generals A (commander), B, C with B loyal, C traitor. Commander sends 'attack'. C tells B 'retreat'. B receives conflicting messages. For  $n = 3m$ , reduce by grouping: simulate each general as  $m + 1$  generals, one traitor in each group to create inconsistency. The  $n \geq 3m + 1$  condition is sufficient via the Oral

Message algorithm OM( $m$ ) recursively sending  $m + 1$  rounds.

**Solution 8:**

Let  $L \in \text{BPP}$  with TM  $M$  using  $m(n)$  random bits. For each input size  $n$ , the set of bad random strings for input  $x$  has size  $\leq 2^{m(n)}/3$ . By union bound over  $2^n$  inputs, the total bad set has size  $\leq 2^n \cdot 2^{m(n)}/3 < 2^{m(n)}$  for  $n$  large enough. So there exists a good random string  $r_n$  that works for all inputs of length  $n$ . Provide  $r_n$  as advice, giving  $\text{BPP} \subseteq \text{P/poly}$ .

**Solution 9:**

For each variable  $x_i$ , create a chain of vertices  $v_{i,1}, \dots, v_{i,k}$  with edges forming a path that can be traversed left-to-right (true) or right-to-left (false). For each clause  $C_j = (\ell_1 \vee \ell_2 \vee \ell_3)$ , insert a clause vertex connected to the corresponding literal vertices in the variable chains. A Hamiltonian cycle exists iff we can pick a consistent direction for each variable chain that hits all clause vertices. This is the standard NP-completeness reduction.

**Solution 10:**

If  $\text{NP} \subseteq \text{P/poly}$ , then SAT has polynomial-size circuits. By the Karp-Lipton theorem,  $\Pi_2^P \subseteq \Sigma_2^P$ , collapsing PH to  $\Sigma_2^P$ . The proof: for any  $\Pi_2^P$  language, the existential player guesses the circuit for SAT and uses it to answer the universal quantifier. This shows  $\text{PH} \subseteq \Sigma_2^P$ , hence  $\text{PH} = \Sigma_2^P$ .

**Solution 11:**

Assume all three hold. During a partition, two nodes cannot communicate. A write to node A must be visible on a read from node B (consistency). But node B must respond (availability) without contacting A (partition). So B has either stale data (no consistency) or must wait (no availability), contradiction. This shows at most two properties can hold simultaneously.

**Solution 12:**

Håstad's switching lemma: a random restriction of a DNF formula leaves a decision tree of depth  $O(\log s)$  with high probability. Apply this to each layer of an  $\text{AC}^0$  circuit of depth  $d$  and size  $S$ . After  $d - 1$  random restrictions, the circuit becomes a decision tree of depth  $O(\log S)^{d-1}$ . Since parity has decision tree depth  $n$ ,  $O(\log S)^{d-1} \geq n$ , so  $S \geq 2^{\Omega(n^{1/(d-1)})}$ .

**Solution 13:**

Nisan's pseudorandom generator for logspace:  $G : \{0,1\}^{O(\log^2 n)} \rightarrow \{0,1\}^n$  fools any logspace machine. The construction uses the INW generator (impagliazzo-Nisan-Wigderson) based on extractors. By enumerating all  $2^{O(\log^2 n)}$  seeds, we can simulate RL in SC (polynomial time, polylog space). This gives  $\text{RL} \subseteq \text{SC} \subseteq \text{P}$ .

**Solution 14:**

$(\supseteq)$  is trivial.  $(\subseteq)$ : Build a PCP verifier for SAT. Start from the NP-complete constraint satisfaction problem with constant degree (via expander graphs). Apply the gap amplification step (repeatedly replace constraints by their AND using composition). The final verifier reads  $O(1)$  proof bits. The theorem shows that approximate solutions to NP

problems remain hard.

**Solution 15:**

If  $P = NP$ , then  $PH = P$  (polynomial hierarchy collapses). Since  $BPP \subseteq \Sigma_2^P \cap \Pi_2^P$  (by Sipser-Gacs theorem), we get  $BPP \subseteq P$ . The inclusion  $BPP \subseteq EXP$  follows from exhaustive simulation of the probabilistic computation over all random bits. Whether  $P = BPP$  is open but believed true due to pseudorandom generators.

**Solution 16:**

$(\Rightarrow)$ : If OWF  $f$  exists, then inverting  $f$  is in  $NP$  (guess a preimage and verify) but not in  $P$  by definition. So  $P \neq NP$ .  $(\Leftarrow)$ : Unconditionally constructing OWF from  $P \neq NP$  is open. However, using Levin's construction, we can define  $f(M, x, 1^t) = (M, x, 1^t)$  where  $M$  outputs the result of  $t$  steps, which is one-way if  $P \neq NP$  under certain assumptions.

**Solution 17:**

$GI \in NP$ : a permutation is a polynomial-time verifiable certificate.  $GI \in coAM$ : the verifier sends a random permutation of one graph; the prover must identify which graph. The interactive protocol has soundness  $1/2$ . If  $GI$  were  $NP$ -complete, then  $PH = \Sigma_2^P$  (Boole'an-Karp'n theorem) because  $coNP \subseteq AM$  and  $AM$  collapses if  $coNP \subseteq AM$ .

**Solution 18:**

Lower bound: construct  $d$  shattered points. The adversary picks a target concept uniformly from  $2^d$  dichotomies. To learn, we must see each point with label to distinguish. By the Sauer-Shelah lemma,  $|\mathcal{H}(m)| \leq (em/d)^d$ . Setting sample size  $m = O(\frac{1}{\epsilon}(d \log(1/\epsilon) + \log(1/\delta)))$  gives the upper bound via the ERM principle.

**Solution 19:**

Progress: if  $\vdash e : \tau$  and  $e$  is closed, then  $e$  is a value or  $e \rightarrow e'$ . Proof by induction on typing derivation. Preservation: if  $\vdash e : \tau$  and  $e \rightarrow e'$ , then  $\vdash e' : \tau$ . Proof by induction on the reduction relation with substitution lemma. For System F, add type abstraction and application cases, with type substitution preserving the subject reduction property.

**Solution 20:**

For a linear  $[n, k]$  code with parity check matrix  $H$ , the minimum distance  $d = \min_{c \neq 0, Hc^T = 0} w(c)$ . Decisional problem (given  $G$  and  $w$ , is  $d \leq w$ ?) is  $NP$ -complete via reduction from 3SAT, by constructing a code whose parity checks encode clause satisfaction. For a linear code over  $\mathbb{F}_q$ , finding minimum weight codeword is  $NP$ -hard.

**Solution 21:**

Define  $\Phi = \text{number of trees} + 2 \cdot \text{number of marked nodes}$ . Insert: add one tree,  $\Delta\Phi = 1$ , actual cost  $O(1)$ , amortized  $O(1)$ . Decrease-key: cut and cascade, each cut adds a tree but may clear marks. Extract-min:  $O(\text{trees} + \text{degree})$  actual cost. By the rank bound  $\text{degree} = O(\log n)$ , and trees decrease by at most  $O(\log n)$ , giving amortized  $O(\log n)$ .

**Solution 22:**

Given set cover instance  $(U, \mathcal{S}, k)$ , construct a directed graph with source  $s$ , set vertices  $s_j$ , element vertices  $u_i$ , edges  $s \rightarrow s_j$  (cost 1),  $s_j \rightarrow u_i$  if  $u_i \in S_j$  (cost 0). Steiner tree

connecting  $s$  to all  $u_i$  uses  $\leq k$  set vertices iff cover of size  $k$  exists. The approximation uses randomized rounding and the junction tree theorem.

**Solution 23:**

Use the moment generating function:  $\Pr[X \geq t] \leq \min_{\lambda > 0} e^{-\lambda t} \mathbb{E}[e^{\lambda X}] = \min_{\lambda > 0} e^{-\lambda t} \prod \mathbb{E}[e^{\lambda X_i}]$ . Since  $\mathbb{E}[e^{\lambda X_i}] = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$ , we get  $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1+\delta)^{1+\delta}})^\mu$  by optimizing  $\lambda = \ln(1 + \delta)$ .

**Solution 24:**

Let  $L \in \text{NP}$  via NTM  $M$  running in time  $p(n)$ . Construct a boolean formula  $\phi$  with variables encoding the computation tableau of  $M$  on input  $x$  of size  $n$ .  $\phi$  has clauses for: (1) initial configuration, (2) valid transitions using  $O(p(n)^3)$  variables, (3) accepting state.  $\phi$  is satisfiable iff  $M$  accepts  $x$ . The reduction runs in polynomial time, proving SAT is NP-hard.

**Solution 25:**

Any comparison tree that sorts  $n$  elements must have at least  $n!$  leaves, so depth  $\geq \log_2(n!) = \Omega(n \log n)$ . Adversary argument: maintain a set of possible permutations consistent with comparisons so far. Each comparison eliminates at most half the remaining permutations, so  $\Omega(\log(n!))$  comparisons needed. For selection, the adversary maintains a partial order ensuring at least  $n - 1$  comparisons are needed for the minimum.